

Viveka-Sengalvarayan / half_add_sub

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Set

0 stars

0 forks

0 watching

Branches

Activity

Tags

Public repository

1 Branch

0 Tags

Go to file

t

Go to file

Add file +

Code

...

 Viveka-Sengalvarayan	Add files via upload	bc6c31c · now	
 db	Add files via upload		now
 incremental_db	Add files via upload		8 minutes ago
 output_files	Add files via upload		7 minutes ago
 simulation/questa	Add files via upload		7 minutes ago
 README.md	Create README.md		18 minutes ago
 c5_pin_model_dump.txt	Add files via upload		17 minutes ago
 half_add_sub.v	Add files via upload		17 minutes ago
 half_adder logic dia.pdf	Add files via upload		17 minutes ago
 half_adder.qpf	Add files via upload		17 minutes ago
 half_adder.qsf	Add files via upload		17 minutes ago
 half_adder.qws	Add files via upload		17 minutes ago
 half_adder.v	Add files via upload		17 minutes ago
 half_sub.qpf	Add files via upload		17 minutes ago
 half_sub.qws	Add files via upload		17 minutes ago
 half_subtractor.qpf	Add files via upload		17 minutes ago
 half_subtractor.qsf	Add files via upload		17 minutes ago
 half_subtractor.v	Add files via upload		17 minutes ago
 wave.sub.pdf	Add files via upload		17 minutes ago



wave.sub.vcd

Add files via upload

17 minutes ago



wave.vcd

Add files via upload

17 minutes ago

HALF_ADDER_SUBTRACTOR

Implementation-of-Half-Adder-and-Half Subtractor-circuit

AIM:

To design a half adder and half subtractor circuit and verify its truth table in Quartus using Verilog programming.

Equipments Required:

Hardware – PCs, Cyclone II , USB flasher

Software – Quartus prime Theory Adders are digital circuits that carry out the addition of numbers.

Half Adder

Half adder is a combinational circuit that performs simple addition of two binary numbers. The input variables designate the augend and addend bits; the output variables produce the sum and carry. It is necessary to specify two output variables because the result may consist of two binary digits.

Sum = $A'B + AB'$ = $A \oplus B$ Carry = AB

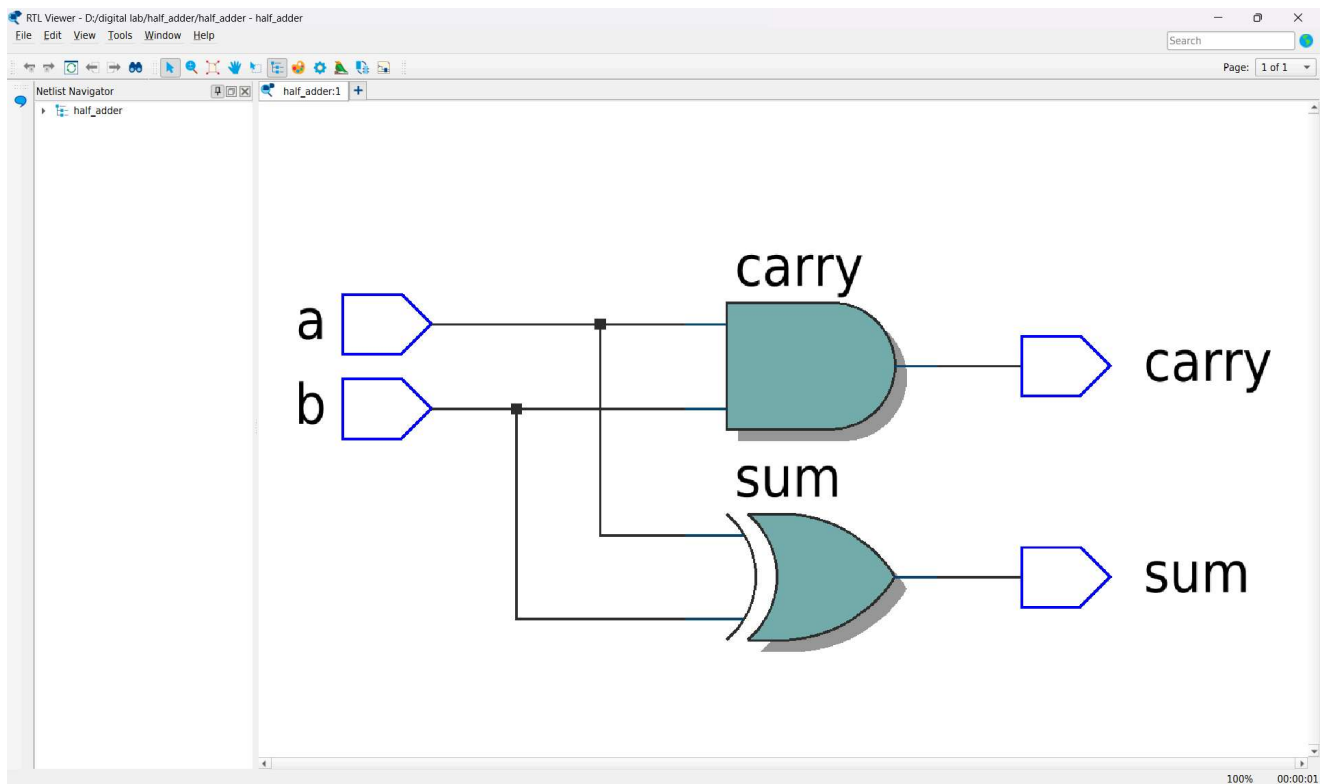
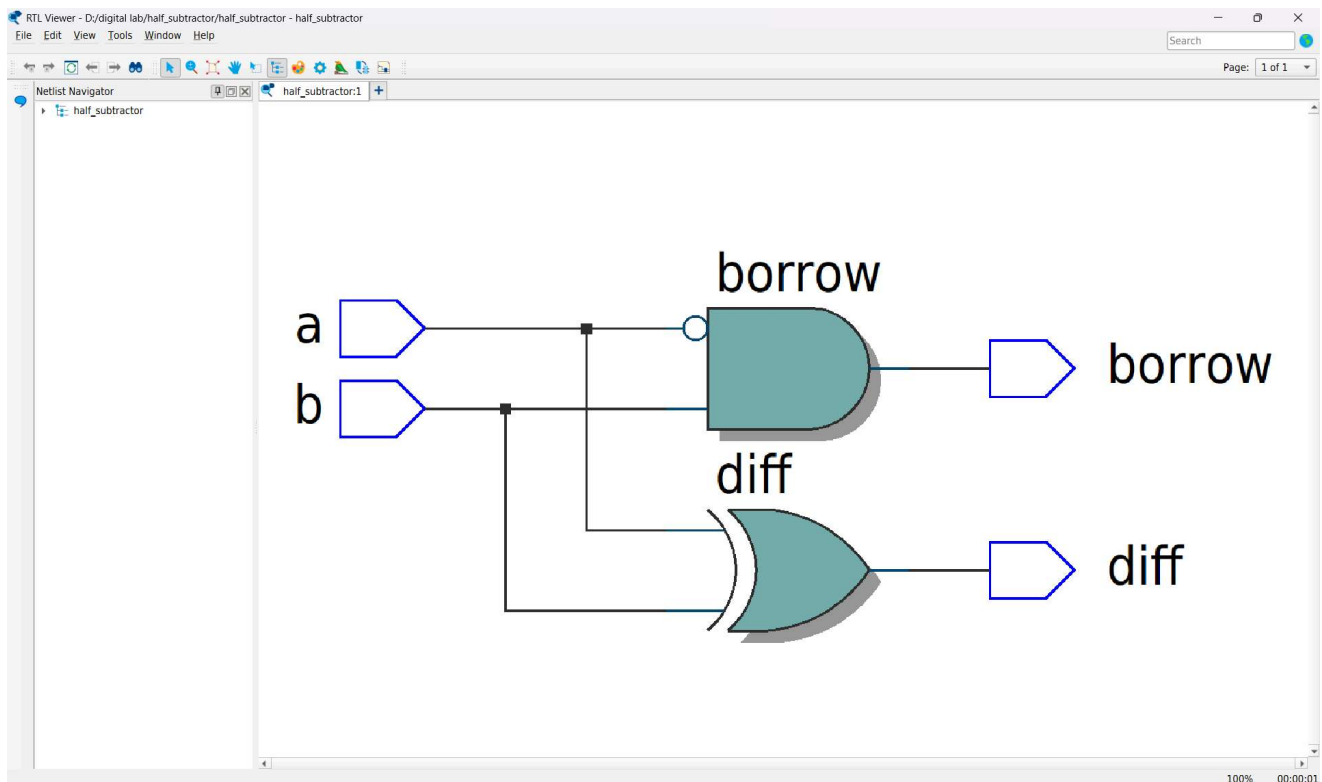


Figure -01 HALF ADDER

Half Subtractor

The half-subtractor is a combinational circuit which is used to perform subtraction of two bits. It has two inputs, X (minuend) and Y (subtrahend) and two outputs D (difference) and B (borrow). To perform $x - y$, we have to check the relative magnitudes of x and y . If $x \geq y$, we have three possibilities: $0 - 0 = 0$, $1 - 0 = 1$, and $1 - 1 = 0$. The result is called the difference bit. If $x < y$, we have $0 - 1$, and it is necessary to borrow a 1 from the next higher stage. The 1 borrowed from the next higher stage adds 2 to the minuend bit, just as in the decimal system a borrow adds 10 to a minuend digit. With the minuend equal to 2, the difference becomes $2 - 1 = 1$. The half-subtractor needs two outputs. One output generates the difference and will be designated by the symbol D. The second output, designated B for borrow, generates the binary signal that informs the next stage that a 1 has been borrowed.

$$\text{Diff} = A'B + AB' = A \oplus B \quad \text{Borrow} = A'B$$



Truthtable

Procedure

1. Type the program in Quartus software.
2. Compile and run the program.
3. Generate the RTL schematic and save the logic diagram.
4. Create nodes for inputs and outputs to generate the timing diagram.
5. For different input combinations generate the timing diagram.

Program:

/* Program to design a half adder and full adder circuit and verify its truth table in quartus using Verilog programming.

// Half Adder in Verilog module half_adder (input wire a, b, // Inputs output wire sum, // Sum output
output wire carry // Carry output);

// Logic equations

assign sum = a ^ b; // XOR for sum

assign carry = a & b; // AND for carry

endmodule

// Half Subtractor in Verilog module half_subtractor (input wire a, b, // Inputs output wire diff, borrow //
Outputs);

```
// Logic equations
assign diff  = a ^ b;    // XOR for difference
assign borrow = ~a & b;  // Borrow when a < b
```

endmodule

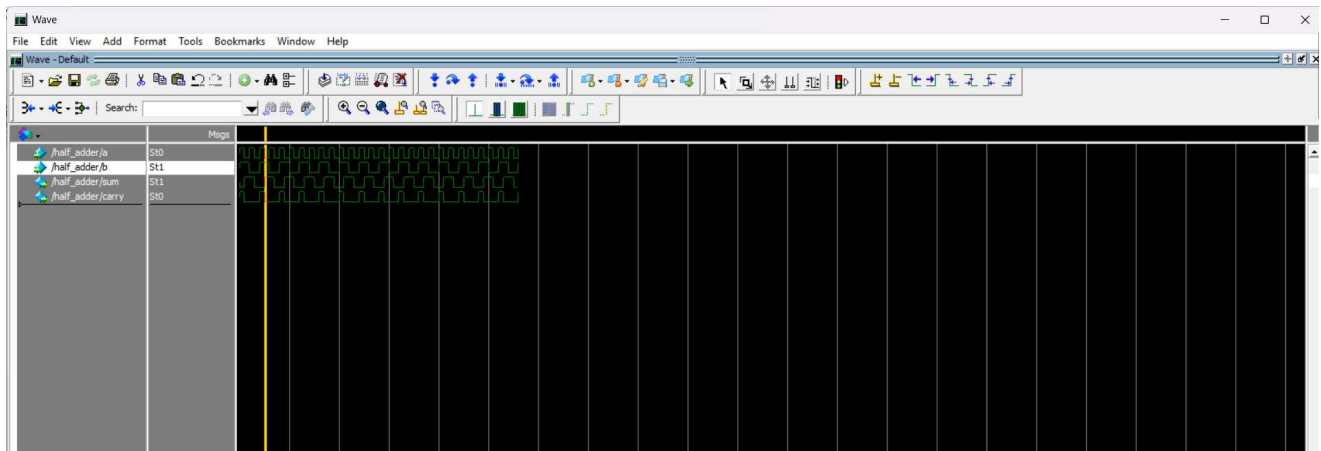
Developed by:viveka

RegisterNumber: 25016820 */

RTL Schematic

Output/TIMING Waveform

half_adder



 README



No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Languages

● Verilog 82.9% ● Standard ML 12.9% ● Scheme 4.2%

Suggested workflows

Based on your tech stack

**SLSA Generic generator**[Configure](#)

Generate SLSA3 provenance for your existing release workflows

**Jekyll using Docker image**[Configure](#)

Package a Jekyll site using the jekyll/builder Docker image.

[More workflows](#)[Dismiss suggestions](#)